

Sample Project Documentation

7.Project Documentation | Credit Card Approval Prediction

Prepared For	Prepared By	Domain	Date
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Abstract

The Credit Card Approval Prediction system applies machine learning to automate the initial screening of credit card applications. The project analyzes applicant financial and demographic details along with credit payment history to predict whether an application is likely to be approved or rejected.

The complete workflow includes dataset collection, data analysis, preprocessing, feature engineering, categorical encoding, model training, model comparison, Flask application integration, and optional IBM Watson Machine Learning deployment.

Problem Statement

Financial institutions need a fast and consistent method for screening large volumes of credit card applications. Manual review is time-consuming and can be affected by inconsistent judgment. This project builds a predictive model to assist with data-driven approval decisions.

Methodology

- Collect applicant and credit history datasets.
- Analyze distributions using univariate, multivariate, and descriptive analysis.
- Clean duplicate and missing records.
- Merge datasets and create model-ready approval/risk labels.
- Train Logistic Regression, Decision Tree, Random Forest, and XGBoost classifiers.
- Evaluate metrics and save the best model.
- Integrate the model with a Flask web application.
- Prepare optional IBM Watson Machine Learning deployment for scalable prediction.

Conclusion

The system demonstrates how machine learning can support faster, more consistent, and scalable credit card approval decisions. It provides hands-on experience in data preprocessing, classification algorithms, Flask deployment, and financial risk assessment.